

Hence $v = k/(x + \text{constant})$

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Answer: A

Current in circuit, $I = P_{\text{supply}}/V_{\text{supply}} = 2400/240 = 10 \text{ A}$

$$P_{\text{kettle}} = P_{\text{supply}} - P_{\text{loss}} = 2400 - (10^2)(0.5+0.5) \\ = 2300 \text{ W}$$

$$\text{Potential difference across kettle} = 240 - (10)(0.5+0.5) \\ = 230 \text{ V}$$

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Answer: B